



Summer Term 2026

Investment and Financial Management
Exercise on Mathematical Basics

Session 04: Annuities

Center for Digital Transformation

Technische Universität München (TUM), Campus Heilbronn

Perpetuity

Idea: The annuity payments are not limited to a certain time horizon ($n \rightarrow \infty$). Although these contracts do not exist frequently, calculations can serve as a good approximation and can be applied for very long investment periods.

Annuity-immediate

$$(1): PV = \lim_{n \rightarrow \infty} C * \frac{1}{q^N} * \frac{q^N - 1}{q - 1} = \lim_{n \rightarrow \infty} C * \frac{1}{q^N} * \left(\frac{q^N}{q - 1} - \frac{1}{q - 1} \right) = \frac{C}{q - 1}$$

Annuity-due

$$(2): PV_{due} = \lim_{n \rightarrow \infty} C * \frac{1}{q^N} * q * \frac{q^N - 1}{q - 1}$$

$$(3): PV_{due} = \lim_{n \rightarrow \infty} C * \frac{1}{q^N} * q * \left(\frac{q^N}{q - 1} - \frac{1}{q - 1} \right) = \frac{q * C}{q - 1}$$

Perpetuity

- A perpetuity is only possible if the annual interest payment is higher than the yearly payout ($C < PV * r$). If this is not the case, the wealth would decrease over time and the payout phase is finite.
- Practical examples (Bonds)
 - Mexico issued in October 2010 and 2011 a bond with 100 years time to maturity, a volume of 1 billion \$ and interest rates of 6.10% and 5.96%.
 - 1752: The finance minister of the UK converted all bonds with fixed maturity into coupons bonds without maturity in order to reduce debt. Debt was repaid in 1923. Since then, there exist another bond with a coupon of 2.5%.

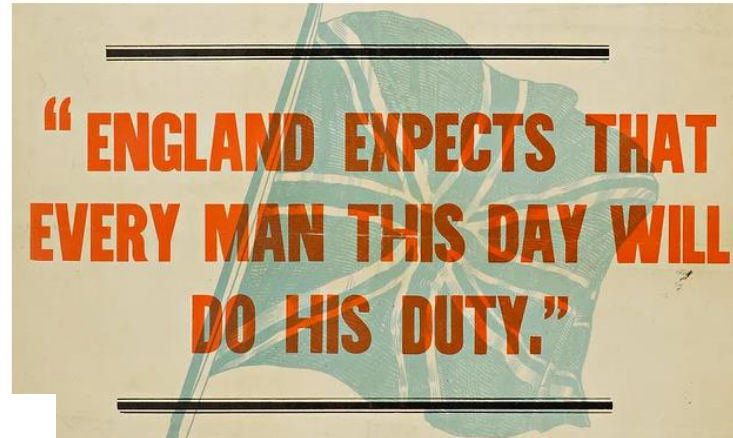
UK bonds that financed first world war to be redeemed 100 years later

Friday 31 October 2014 10.53 GMT

As Europe marks the centenary of the Great War, the Treasury said it would pay off £218m from a 4% consolidated loan next February, as part of a redemption of bonds stretching as far back as the 18th century. They also relate to the South Sea Bubble crisis of 1720, the Napoleonic and Crimean wars and the Irish potato famine.

The national war bonds, which paid out a rate of interest of 5%, were issued in 1917 as the government sought to raise more money to finance the ongoing cost of the first world war, which started with the issue of the first war loan in November 1914. The bonds were sold to private investors in 1917 with the advertisement: "If you cannot fight, you can help your country by investing all you can in 5 per cent Exchequer Bonds ... Unlike the soldier, the investor runs no risk."

In addition to the war bonds, some of the debt being repaid dates as far back as the eighteenth century. In 1853, William Gladstone, then chancellor, consolidated the capital stock of the South Sea Company which had collapsed in the infamous South Sea Bubble financial crisis of 1720. And in 1888, then chancellor, George Goschen, converted bonds first issued in 1752 which were later used to finance the Napoleonic and Crimean wars, the Slavery Abolition Act (1835) and the Irish Distress Loan (1847). This debt will be repaid through the redemption of the 4% consols.



- 2.1 Introduction**
- 2.2 Application of annuity calculation**
- 2.3 Constant annuities**
 - 2.3.1 Yearly annuities**
 - 2.3.2 Intra-year annuities**
- 2.4 Varying annuities**
 - 2.4.1 Arithmetic progression**
 - 2.4.2 Geometric progression**

Intra-year annuities

We differentiate between three cases

1. Annuity period = Interest period
2. Annuity payments are made intra-yearly, interest payments annually
(more than one annuity payments within one interest period)
3. Annuity payments are made annually, interest payments intra-yearly
(more than one interest payments within one annuity period)

Case 1: Annuity period equals interest period

- **Annuity period and interest period are identical**
 - If intra-year interest rate $\frac{r}{m}$ and maturity $n * m$ is given in terms of the corresponding period (e.g. quarters or months instead of years), there are no changes compared to the calculation with annual periods.

For annuities-immediate, it holds:

$$(1): FV = C * \frac{q^{n*m} - 1}{q - 1}, \text{ with } q = 1 + \frac{r}{m}$$

$$(2): PV = \frac{FV}{q^{n*m}} = \frac{q^{n*m} - 1}{q^{n*m+1} - q^{n*m}} * C$$

Replacing r by $\frac{r}{m}$ and N by $n * m$ is done analogous for annuities-due.

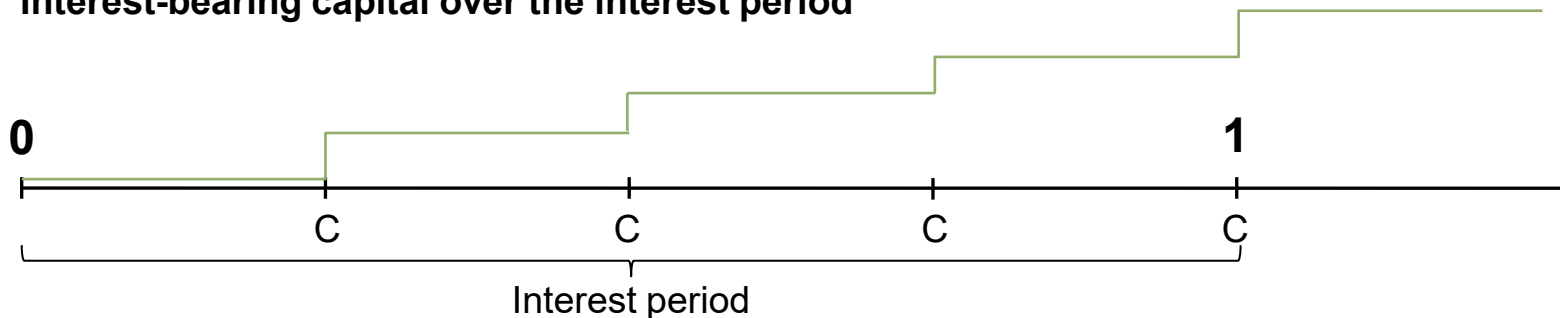
(A.9) Intra-year annuities

It should be invested in a monthly annuity-immediate of 1,000 EUR. The monthly interest rate equals 0.33% ($\frac{r}{m} = \frac{0.04}{12}$). The time to maturity equals 10 years. Calculate the future value of this annuity.

Case 2: Annuity payments are made intra-yearly, interest payments annually

- **During an interest period more than one annuity payments occur**, e.g. yearly calculated interest but monthly annuity: mortgage credit or private small loans
- How do we deal with this case?
 - **Problem:** The committed capital is **not constant** over the interest period.
 - We construct a **fictitious periodic annuity** with the same time structure as the interest.
 - Then, we can use the known formulas for annuity calculation.

Interest-bearing capital over the interest period



Case 3: Interest payments are made intra-yearly, annuity payments annually

Idea: Annuity period is longer than the interest period, e.g. quarterly interest payments, yearly annuity payments

How do we deal with this case?

- Comparable approach as before: Construction of **fictitious periodic interest rate r' (effective interest rate)**

$$(1): r' = \left(\left(1 + \frac{r}{m} \right)^m - 1 \right) \Rightarrow r' = e^r - 1 \text{ for continuous compounding } (m \rightarrow \infty)$$

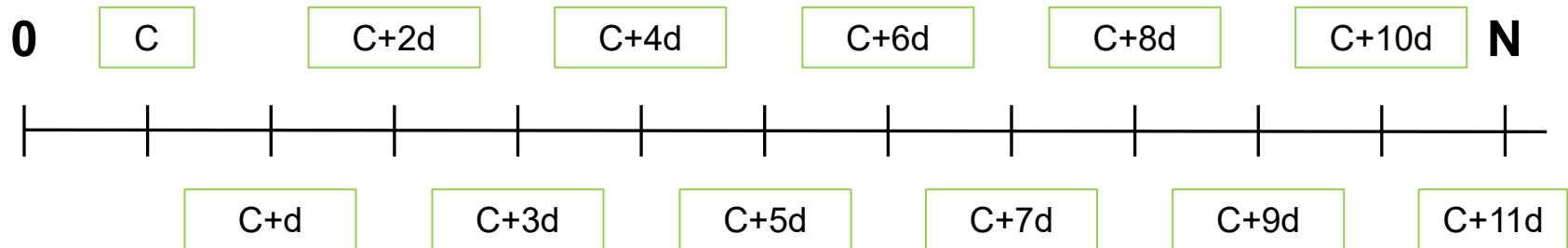
Now you can apply the known annuity formulas using the fictitious period interest rate r' .

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Varying annuities with arithmetic progression

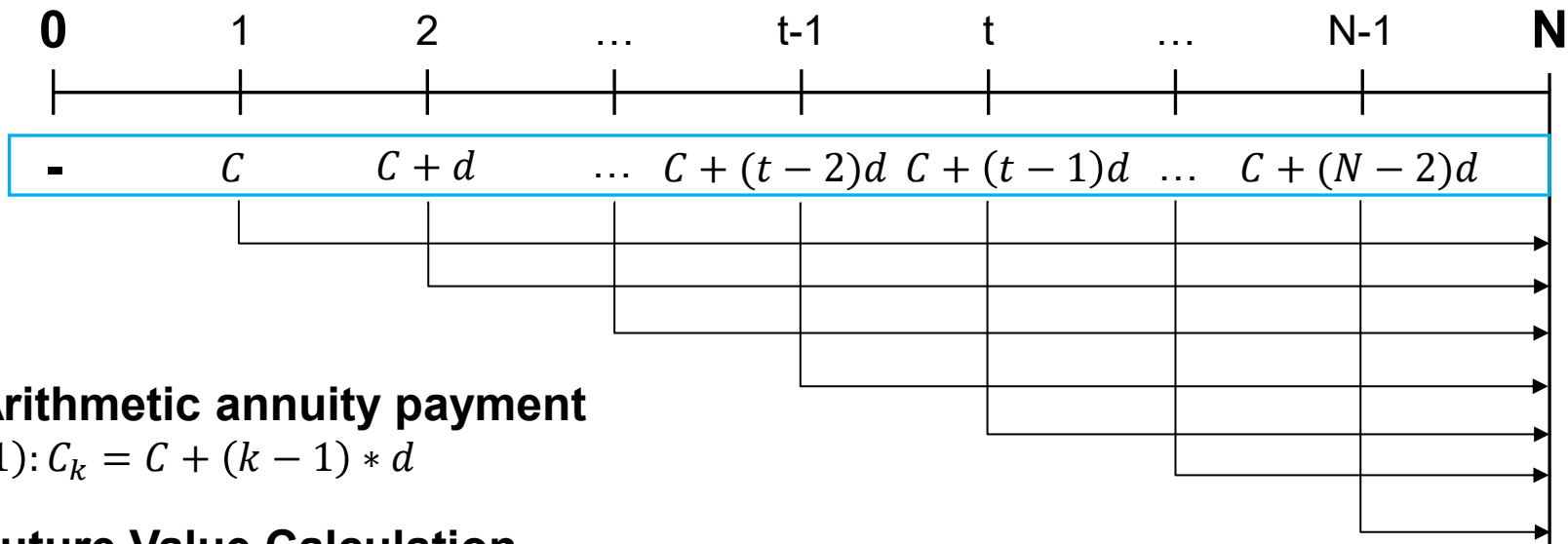
Idea: The annuity increases or decreases by a constant amount d



For $d < 0$ or $d > 0$, it results an arithmetic numeric series consisting of...

(1): $C, C + d, C + 2d, C + 3d, C + 4d, \dots, C + (N - 1)d$

Future Value of an annuity-immediate with arithmetic progression



Arithmetic annuity payment

(1): $C_k = C + (k-1) * d$

Future Value Calculation

(2): General formula: $(C + (k-1) * d) * q^{N-k}$

(3): $FV = C * q^{N-1} + (C + d) * q^{N-2} + \dots + (C + (N-2) * d) * q + C + (N-1) * d$

Future Value of an annuity-immediate and arithmetic progression

$$(1): FV = C * q^{N-1} + (C + d) * q^{N-2} + \dots + (C + (N - 2) * d) * q + C + (N - 1) * d$$

$$(2): FV = \sum_{k=1}^N (C + (k - 1) * d) * q^{N-k} = C * \underbrace{\sum_{k=1}^N q^{N-k}}_{FVF} + d * \underbrace{\sum_{k=1}^N (k - 1) * q^{N-k}}_{:= x}$$

$$(3): x = \sum_{k=1}^N (k - 1) * q^{N-k} = 0 * q^{N-1} + 1 * q^{N-2} + \dots + (N - 1) * q^0$$

$$(4): q * x = 1 * q^{N-1} + 2 * q^{N-2} + \dots + (N - 1) * q^1$$

$$(5): qx - x = q^{N-1} + (2q^{N-2} - 1q^{N-2}) + \dots = q^{N-1} + q^{N-2} + q^{N-3} + \dots + q^1 - (N - 1)$$

$$(6): qx - x = 1 + q^1 + q^2 + \dots + q^{N-1} - N$$

$$(7): qx - x = \frac{q^N - 1}{q - 1} - N \Leftrightarrow x = \frac{1}{q - 1} * \left(\frac{q^N - 1}{q - 1} - N \right)$$

$$(8): FV = C * \frac{q^N - 1}{q - 1} + \frac{d}{q - 1} * \left(\frac{q^N - 1}{q - 1} - N \right)$$

$$(9): FV = C * FVF + \frac{d}{q - 1} * (FVF - N)$$

- The Future Value of a number N of annuity-immediate payments with arithmetic progression $C + (k - 1) * d$ can be easily calculated as:

$$(1): FV = \left(C + \frac{d}{q - 1} \right) * \frac{q^N - 1}{q - 1} - \frac{N * d}{q - 1}$$

- The Present Value of a number N of annuity-immediate payments with arithmetic progression equals:

$$(2): PV = \frac{FV}{q^N} = \left(C + \frac{d}{q - 1} \right) * \frac{q^N - 1}{q^N * (q - 1)} - \frac{N * d}{q^N * (q - 1)}$$

(A.10) Varying annuities

Annuities of 1,000 EUR are paid at the end of every month for the next 10 years. The annuity payments increase by an amount $d = 5$ EUR every month. Interests are calculated monthly and rates remain at a level of 0.4%. What will be the Present Value and Future Value of this annuity?

(A.11) Varying annuities

At the end of each of the following five years, annuity payments should be paid. The annuity payments increase from 10,000 EUR in the first year by 1,000 EUR each year. What is the required initial capital if the interest rate is assumed to be 5% p.a.?

$$A = 42,123.43 \text{ EUR}$$

$$B = 45,987.23 \text{ EUR}$$

$$C = 51,531.68 \text{ EUR}$$

$$D = 55,122.24 \text{ EUR}$$

- The Future Value of a number N of **annuity-due** payments with arithmetic progression $C + (k - 1) * d$ can be easily calculated as:

$$(1): FV_{due} = q * FV = q * \left(C + \frac{d}{q - 1} \right) * \frac{q^N - 1}{q - 1} - \frac{N * d * q}{q - 1} = q * C * FVF + \frac{q * d}{q - 1} * (FVF - N)$$

$$(2): FV_{due} = C_{due} * FVF_{due} + \frac{d}{q - 1} * (FVF_{due} - N * q)$$

- The Present Value of a number N of **annuity-due** payments with arithmetic progression equals:

$$(2): PV_{due} = q * PV = \frac{FV_{due}}{q^N} = \left(C_{due} + \frac{d}{q - 1} \right) * \frac{q^N - 1}{q^{N-1} * (q - 1)} - \frac{N * d}{q^{N-1} * (q - 1)}$$

$$(3): PV_{due} = C_{due} * PVF_{due} + \frac{d}{q - 1} * \left(PVF_{due} - \frac{N}{q^{N-1}} \right)$$

(A.12) Varying annuities

At the beginning of each of the following five years, annuity payments should be paid. The annuity payments increase from 10,000 EUR in the first year by 1,000 EUR each year. What is the required initial capital if the interest rate is assumed to be 5% p.a.?

(A.13) Varying annuities

An annual annuity amount of 20,000 EUR should go up every year by 200 EUR. What is the Present Value of the annuity if interest rates are assumed to be fixed at 6.5% and the investment horizon is 15 years?

- (a) Please consider annuity-immediate calculation.**
- (b) Please consider annuity-due calculation.*
- (c) What is the total amount of annuities which is paid without interests?*

(A.13) Varying annuities

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(b) Please consider annuity-due calculation.

(c) What is the total amount of annuities which is paid without interests?

(A.13) Varying annuities

An annual annuity amount of 20,000 EUR should go up every year by 200 EUR. What is the Present Value of the annuity if interest rates are assumed to be fixed at 6.5% and the investment horizon is 15 years?

(a) Please consider annuity-immediate calculation.

(b) Please consider annuity-due calculation.

(c) What is the total amount of annuities which is paid without interests?

1. Future Value FV

$$FV = \left(C + \frac{d}{q - 1} \right) * \frac{q^N - 1}{q - 1} - \frac{N * d}{q - 1}$$

2. Present Value PV

$$PV = \left(C + \frac{d}{q - 1} \right) * \frac{q^N - 1}{q^N * (q - 1)} - \frac{N * d}{q^N * (q - 1)}$$

3. Initial Annuity C

$$C = \left(FV + \frac{N * d}{q - 1} \right) * \frac{q - 1}{q^N - 1} - \frac{d}{q - 1}$$

$$C_k = C + (k - 1) * d$$

4. Investment Period N

Equations cannot be explicitly solved for N .

1. Future Value FV

$$FV_{due} = q * \left(C + \frac{d}{q-1} \right) * \frac{q^N - 1}{q-1} - \frac{N * d * q}{q-1}$$

2. Present Value PV

$$PV_{due} = \left(C_{due} + \frac{d}{q-1} \right) * \frac{q^N - 1}{q^{N-1} * (q-1)} - \frac{N * d}{q^{N-1} * (q-1)}$$

3. Initial Annuity C

$$C = \left(FV + \frac{N * d * q}{q-1} \right) * \frac{q-1}{q * (q^N - 1)} - \frac{d}{q-1}$$

$$C_{due,k} = C_{due} + (k-1) * d$$

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Varying annuities with geometric progression

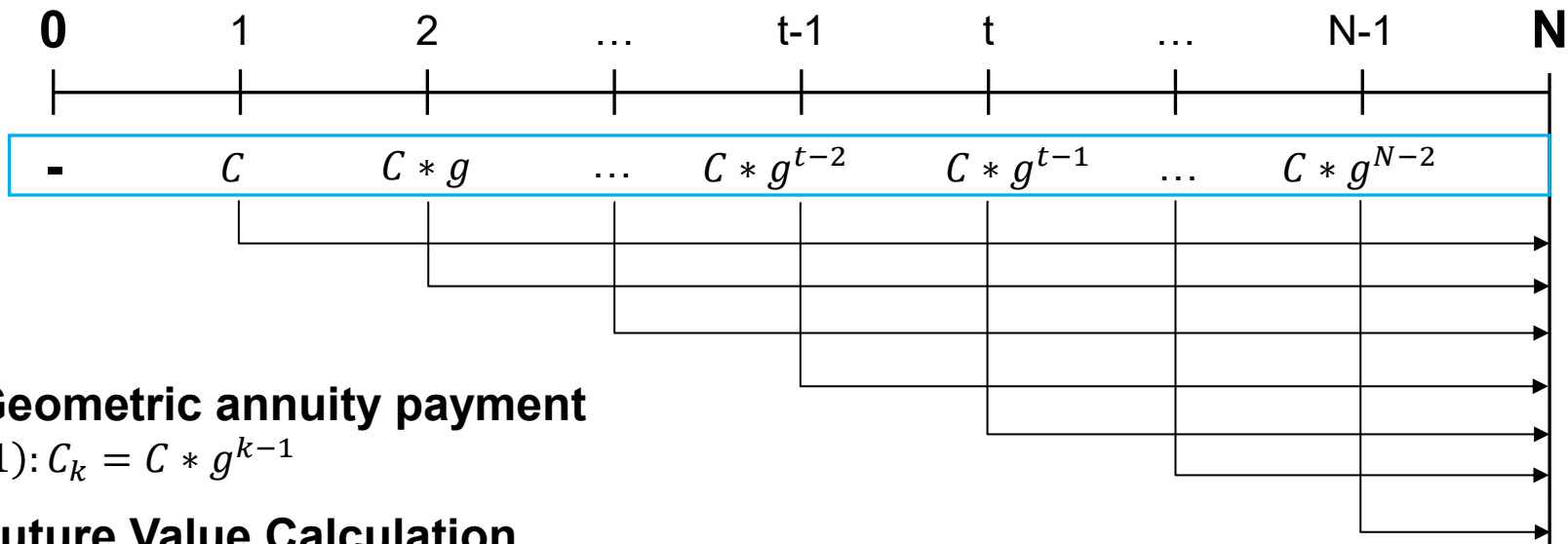
By using geometrically progressing annuities, a company can model the development of certain factors to be able to determine the appropriate amount of pension provisions.

- Inflation
- Wages
- Capital market risk
- Interest rates
- Investments

These developments cannot be modeled by arithmetically progressing annuities.

Idea: Let the annuities progress by a constant factor g !

Future Value of an annuity-immediate with arithmetic progression



Geometric annuity payment

(1): $C_k = C * g^{k-1}$

Future Value Calculation

(2): General formula: $(C * g^{k-1}) * q^{N-k}$

(3): $FV = C * q^{N-1} + (C * g) * q^{N-2} + \dots + C * g^{N-2} * q + C * g^{N-1} * d$

Varying annuities with geometric progression

Idea: Let the annuities progress by a constant factor g

- Adjusting the general annuity formula to the geometric progressing annuity payment:

$$(1): FV = \sum_{k=1}^N C_k * q^{N-k}; \text{ with } C_k = C * g^{k-1}$$

$$(2): FV = \sum_{k=1}^N C * g^{k-1} * q^{N-k}$$

Future Value with geometric progression: We need to differentiate two cases.

$$(F): FV = \sum_{k=1}^N C * g^{k-1} * q^{N-k}$$

Case I: $g \neq q$

$$(1): FV = C * \frac{q^N}{g} * \sum_{k=1}^N (g^k * q^{-k}) = C * \frac{q^N}{g} * \sum_{k=1}^N \left(\frac{g}{q}\right)^k = C * \frac{q^N}{g} * \left(\frac{g}{q}\right) * \sum_{k=1}^N \left(\frac{g}{q}\right)^{k-1}$$

$$(2): FV = C * \frac{q^N}{g} * \frac{g}{q} * \frac{\left(\frac{g}{q}\right)^N - 1}{\left(\frac{g}{q}\right) - 1} = C * \frac{g^N - q^N}{g - q}$$

Future Value with geometric progression: We need to differentiate two cases.

$$(F): FV = \sum_{k=1}^N C * g^{k-1} * q^{N-k}$$

Case II: $g = q$

$$(1): FV = C * \frac{q^N}{g} * \sum_{k=1}^N (g^k * q^{-k}) = C * \frac{q^N}{g} * \sum_{k=1}^N \left(\frac{g}{q}\right)^k$$

$$(2): FV = C * \frac{q^N}{g} * N = C * q^{N-1} * N$$

- The Future Value of a number N of **annuity-immediate** payments with geometric progression $C * g^{k-1}$ can be calculated as:

$$(1): FV = \begin{cases} C * N * q^{N-1} & : \text{if } g = q \\ C * \frac{g^N - q^N}{g - q} & : \text{if } g \neq q \end{cases}$$

- The Present Value of a number N of **annuity-immediate** payments with geometric progression $R * g^{k-1}$ can be calculated as:

$$(2): PV = \frac{FV}{q^N} = \begin{cases} \frac{C * N}{q} & : \text{if } g = q \\ C * \frac{\left(\frac{g}{q}\right)^N - 1}{g - q} & : \text{if } g \neq q \end{cases}$$

- The Future Value of a number N of **annuity-due** payments with geometric progression $C * g^{k-1}$ can be calculated as:

$$(1): FV_{due} = \begin{cases} C_{due} * N * q^N & : \text{if } g = q \\ C_{due} * q * \frac{g^N - q^N}{g - q} & : \text{if } g \neq q \end{cases}$$

- The Present Value of a number N of **annuity-due** payments with geometric progression $R * g^{k-1}$ can be calculated as:

$$(2): PV_{due} = \begin{cases} C_{due} * N & : \text{if } g = q \\ C_{due} * q * \frac{\left(\frac{g}{q}\right)^N - 1}{g - q} & : \text{if } g \neq q \end{cases}$$

Solving for the investment period

Annuities-immediate

$$(1): N = \begin{cases} \frac{PV * q}{C} & : \text{if } g = q \\ \frac{\ln\left(1 + \frac{PV * (g - q)}{C}\right)}{\ln\left(\frac{g}{q}\right)} & : \text{if } g \neq q \end{cases}$$

Annuities-due

$$(2): N = \begin{cases} \frac{PV_{due}}{C_{due}} & : \text{if } g = q \\ \frac{\ln\left(1 + \frac{PV_{due} * (g - q)}{C_{due} * q}\right)}{\ln\left(\frac{g}{q}\right)} & : \text{if } g \neq q \end{cases}$$

1. Future Value FV

$$FV = \begin{cases} C * N * q^{N-1} & : \text{if } g = q \\ C * \frac{g^N - q^N}{g - q} & : \text{if } g \neq q \end{cases}$$

2. Present Value PV

$$PV = \frac{FV}{q^N} = \begin{cases} \frac{C * N}{q} & : \text{if } g = q \\ C * \frac{\left(\frac{g}{q}\right)^N - 1}{g - q} & : \text{if } g \neq q \end{cases}$$

3. Initial Annuity C

$$C = \begin{cases} \frac{BW * q}{N} & : \text{if } g = q \\ BW * \frac{g - q}{\left(\frac{g}{q}\right)^N - 1} & : \text{if } g \neq q \end{cases}$$

4. Investment Period N

$$N = \begin{cases} \frac{PV * q}{C} & : \text{if } g = q \\ \frac{\ln\left(1 + \frac{PV * (g - q)}{C}\right)}{\ln\left(\frac{g}{q}\right)} & : \text{if } g \neq q \end{cases}$$

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$$FV_{due} = \begin{cases} C_{due} * N * q^N & : \text{if } g = q \\ C_{due} * q * \frac{g^N - q^N}{g - q} & : \text{if } g \neq q \end{cases}$$

2. Present Value PV

$$PV_{due} = \begin{cases} C_{due} * N & : \text{if } g = q \\ C_{due} * q * \frac{\left(\frac{g}{q}\right)^N - 1}{g - q} & : \text{if } g \neq q \end{cases}$$

3. Initial Annuity C

$$C = \begin{cases} \frac{PV_{due}}{N} & : \text{if } g = q \\ PV_{due} * \frac{g - q}{q * \left(\left(\frac{g}{q}\right)^N - 1\right)} & : \text{if } g \neq q \end{cases}$$

$$C_{due,k} = C_{due} * g^{k-1}$$

4. Investment Period N

$$N = \begin{cases} \frac{PV_{due}}{C_{due}} & : \text{if } g = q \\ \frac{\ln\left(1 + \frac{PV_{due} * (g - q)}{C_{due} * q}\right)}{\ln\left(\frac{g}{q}\right)} & : \text{if } g \neq q \end{cases}$$

(A.14) Exemplary exercises

A craftsman sells his small craft business with a current value of 300,000 EUR due to succession problems to an investor. The craftsman intends to receive an annual annuity of 20,000 EUR to be able to cover his expenses during his retirement. The annual payment should increase by 4% each year. How long can the annuity be paid if we assume a constant interest rate of 5%?

(a) Annuity-immediate?

(b) Annuity-due?

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(a) Annuity-immediate?

(b) Annuity-due?

(A.15) Exemplary exercises

A company wants to pay its former manager an annuity-immediate of 15,000 EUR which increases by 5% each year. What is the amount of capital which need to be initially invested so that the annuity can be paid to the manager for 10 years if interest rates are fixed at (a) 5% and (b) 6%. Please also calculate the sum of all annuity payments.

(A.16) Exemplary exercises

As price for selling his house a seller gets the following annuity promise. He gets an annual annuity-immediate of 24,000 EUR for 10 years and may choose between the following conditions. Which condition should the seller take if interest rates are fixed at 5% p.a.?

- (a) Constant increase by 1,000 EUR each year*
- (b) Increase by 6% each year*

A = Option (a)

B = Option (b)

C = Indifferent

*D = Cannot be answered
with given information*

1

2

3